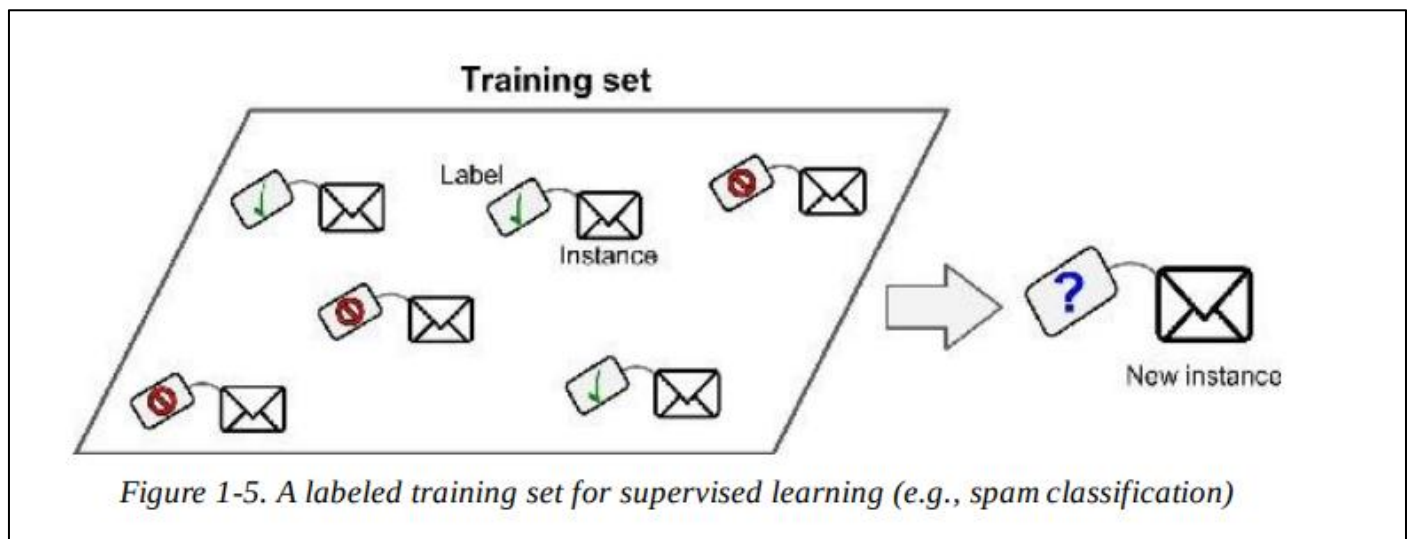


Machine Learning: Supervised Learning

Machine Learning systems can be classified according to the amount and type of supervision they get during training. There are four major categories: supervised learning, unsupervised learning, semisupervised learning, and Reinforcement Learning.

Supervised learning

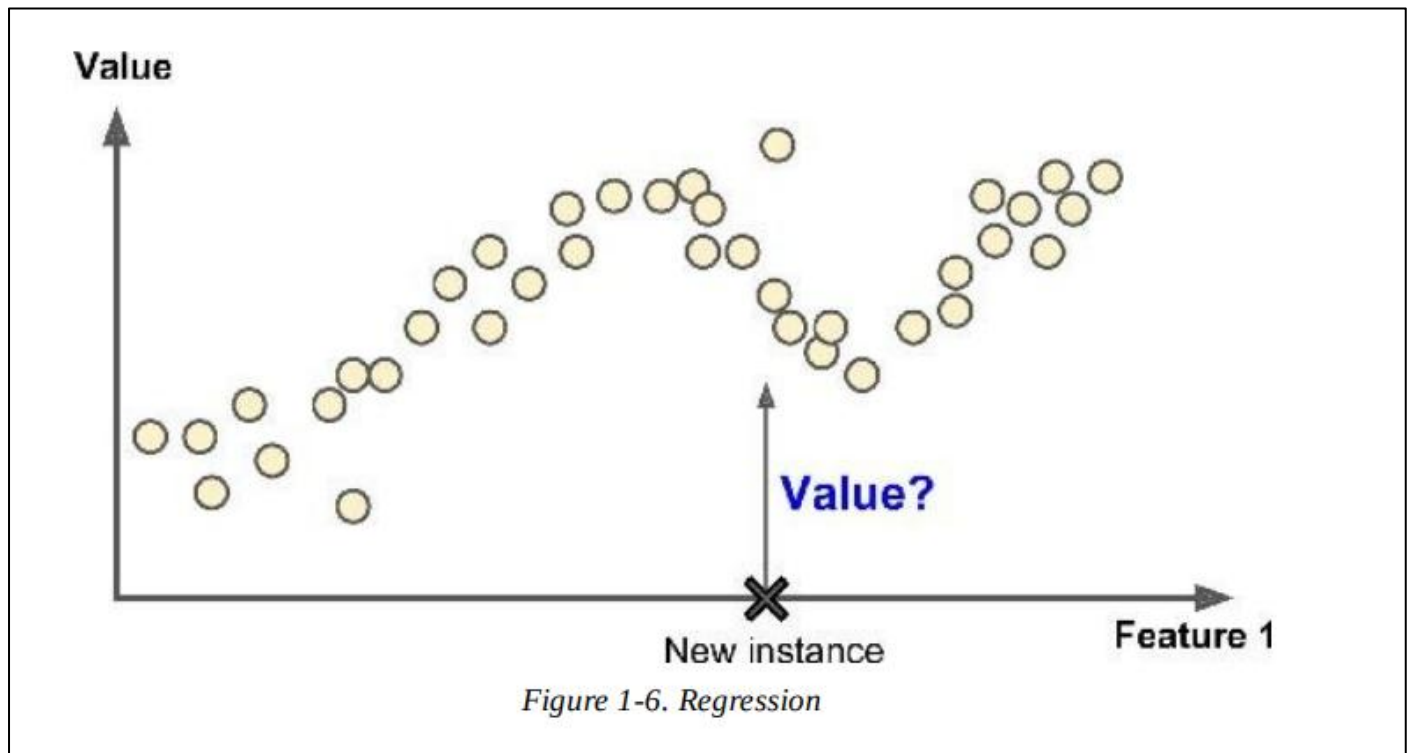
In supervised learning, the training data you feed to the algorithm includes the desired solutions, called labels (Figure 1-5).



A labeled training set for supervised learning (e.g., spam classification) A typical supervised learning task is classification. The spam filter is a good example of this: it is trained with many example emails along with their class (spam or ham), and it must learn how to classify new emails.

Another typical task is to predict a target numeric value, such as the price of a car, given a set of features (mileage, age, brand, etc.) called predictors. This sort of task is called regression (Figure 1-6). 1 To train the system, you need to give it many examples of cars, including both their predictors and their labels (i.e., their prices).

NOTE In Machine Learning an attribute is a data type (e.g., “Mileage”), while a feature has several meanings depending on the context, but generally means an attribute plus its value (e.g., “Mileage = 15,000”). Many people use the words attribute and feature interchangeably, though.



Note that some regression algorithms can be used for classification as well, and vice versa. For example, Logistic Regression is commonly used for classification, as it can output a value that corresponds to the probability of belonging to a given class (e.g., 20% chance of being spam).

Here are some of the most important supervised learning algorithms:

- K-Nearest Neighbors
- Linear Regression
- Logistic Regression
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural networks

Source: [Hands-On Machine Learning with Scikit-Learn and TensorFlow](#)